

Tyson Gentry

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SUMMARY

Computer Science student (expected December 2027) with strong foundations in algorithms, Python development, SQL, and Linux environments. Experienced in AI-assisted software engineering using modern LLMs such as GPT-5 and passionate about scalable AI systems and performance-focused development. Seeking a Summer 2026 Software Engineering Internship.

EDUCATION

University of Houston, College of Natural Sciences and Mathematics Bachelor of Science in Computer Science, Relevant Coursework: Data Structures, Algorithms, Software Engineering, Database Systems, AI/ML Cedar Valley College, College of Science Associates of Science	Houston, TX December 2027 Lancaster, TX May 2024
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Technical Skills

- Languages: Python, C++, JavaScript, Swift
- Libraries/Frameworks: Pandas, NumPy, React
- Tools: Git, GitHub, Linux, Excel, PowerPoint
- Concepts: Data Structures & Algorithms, Computer Security, Computer Organization & Architecture

Projects

Shortest Path Graph Solver (C++)

- Engineered a high-performance graph solver using Dijkstra's algorithm in C++, optimizing for $O(E \log V)$ time complexity using priority queues.
- Validated system accuracy by using a Linux-based testing suite to process and verify complex weighted graph inputs.

Knight's Tour Backtracking Solver (C++)

- Developed a recursive backtracking algorithm to solve closed Knight's Tour problems.
- Implemented board constraints, blocked tiles, and formatted dynamic output.

Data Pipeline Automation (Python)

- Architected a data preprocessing pipeline that automated the cleaning and normalization of large-scale CSV datasets, reducing manual preparation time by an estimated 50%.
- Implemented data validation checks using NumPy to identify and handle null values and outliers, ensuring high-quality inputs for downstream analysis.
- Automated data preparation to improve usability for analysis workflows.

Portfolio Website (HTML & CSS) — In Progress

- Building a responsive portfolio website using HTML, CSS, and React to showcase software projects and technical experience.
- Leveraging Cursor AI for AI-assisted development, including code generation, refactoring, debugging, and design iteration to improve development efficiency and code quality.

Experience

Walmart – Associate	Desoto, TX
• Managed and audited inventory data for high-volume retail operations, ensuring 100% data integrity for thousands of SKUs via proprietary scanning software. • Troubleshoot technical inconsistencies in scanning hardware/software, reducing processing delays and improving workflow efficiency.	

Leadership & Involvement

• National Society of Black Engineers – Member	(Aug 2024 – Present)
• Black Men in STEM	(Aug 2025 – Present)
• Collegiate 100 – Member	(Dec 2024 – Present)
• ColorStack – Member	(Aug 2025 – Present)
• Black Male Engineering Network	(Dec 2024 – Present)